Qinxun Li

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Education

2023– **Ph.D. Candidate in Physics (Astronomy Track)**
Department of Physics and Astronomy, **University of Utah**
Research Advisor: Prof. Kyle Dawson
Research Topic: Geometric Probes of Cosmic Acceleration from Galaxy Surveys

2019–2023 **Bachelor of Science in Astronomy**
Department of Astronomy, **University of Science and Technology of China (USTC)**
Research Advisor: Prof. Huiyuan Wang & Prof. Wentao Luo
Thesis: *Constraining Satellite Galaxy–Subhalo Connection with Weak Gravitational Lensing*

Research Leadership

2026– **Co-lead, BAO Topical Group**, DESI Collaboration (DR3 analysis)
Science: The DESI DR2 BAO measurements, combined with CMB and supernova data, showed a 2.8–4.2σ preference for evolving dark energy over a cosmological constant. DR3 extends the DESI map with additional years of observations. Its BAO analysis will deliver the most precise distance measurements to date and test whether this preference persists.
Role: Set the analysis plan and timeline, assign projects across the working group, and make key decisions with the other co-lead and the collaboration leadership. Will lead the pipeline validation and coordinate the writing of the DR3 BAO key paper.

Publications

Metrics: *Total (first-author) number:* ***32(4)*** *Total (first-author) citations:* ***3239(27)*** *H-index:* ***13.***
Link to an updated list of my publications: ui.adsabs.harvard.edu/user/libraries/i25FBgobQbKAeoJoK_SaRQ

First-Author Papers

1. **Li, Q.** et al. "DESI DR2 GLAM-Uchuu covariance mocks: BAO Validation and Measurements with Mock-based Covariance Matrices" (2026). In collaboration-wide review; expected to be submitted in early December 2026.
2. **Li, Q.** and S. Liu. "Implications of Inelastic Dark Matter for Primordial Dark-Star Evolution: Kinematic Thresholds and Nonthermal Capture". arXiv:2609.10662 [hep-ph] (2026).
3. **Li, Q.** "Kinematic Lensing Ratio: Reviving Weak Lensing Cosmography as a Geometric Dark Energy Probe". arXiv:2604.26015 [astro-ph.CO] (2026). Resubmitted to PRL after addressing the first-round referee comments.
4. **Li, Q.**, M. Kilbinger, W. Luo, K. Wang, H. Wang, et al. "Black Hole–Halo Mass Relation from UNIONS Weak Lensing". *ApJL* 969, L25 (2024).

DESI Papers with Major Contributions

1. DESI Collaboration, A. G. Adame, J. Aguilar, S. Ahlen, O. Alves, et al. (including **Q. Li**). "DESI DR2 Results V: The Physical Properties of the DESI Cosmological Tracer Populations" (2026). In collaboration-wide review; expected to be submitted in early December 2026.

   My contribution: *I mentored an undergraduate student, Hanzhang Zhang, to perform the completeness analysis of emission line galaxies. Also, I analyzed the target selection completeness of DESI luminous red galaxies and wrote the corresponding sections in the paper. The authorship of this paper is alphabetical.*

2. Zhao, C., **Q. Li**, et al. "The galaxy-halo connection of DESI extreme emission line galaxies at $z < 0.9$" (2026). In collaboration-wide review; expected to be submitted in early December 2026.

   My contribution: *I build the galaxy samples for the whole analysis and perform the clustering measurements as the input for the HOD modeling.*

3. Andrade, U., E. Paillas, J. Mena-Fernández, **Q. Li**, A. J. Ross, et al. "Validation of the DESI DR2 measurements of baryon acoustic oscillations from galaxies and quasars". *Phys. Rev. D* 112, 083512 (2025).

   My contribution: *I performed the intrinsic galaxy property split tests and several post-unblinding tests described in Secs. VII.B and VII.D, including the DESI LRG2 vs BOSS CMASS + eBOSS LRG comparison, which settled down a popular concern in the community.*

4. DESI Collaboration, M. Abdul Karim, J. Aguilar, S. Ahlen, S. Alam, et al. (including **Q. Li**). "DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints". *Phys. Rev. D* 112, 083515 (2025).

   My contribution: *I performed the consistency tests of BAO measurements (DESI DR1 vs DR2 and SDSS vs DESI) and wrote Section III.C. I also made Figures 6 and 13, which show the BAO and SN distance measurements in comparison to various cosmological models. The authorship of this paper is alphabetical.*

Other Contributing Papers

1. Fernandez-Garcia, E., F. Prada, A. Klypin, A. Ross, Z. Ding, et al. (including **Q. Li**). "DESI DR2 GLAM-Uchuu covariance mocks: Clustering results from BGS-BRIGHT and LRGs" (2026). In collaboration-wide review; expected to be submitted in early December 2026.

2. Vaisakh, R., N. K. Khan, A. Klypin, R. Kehoe, E. Fernandez-Garcia, et al. (including **Q. Li**). "DESI DR2 GLAM-Uchuu covariance mocks: Clustering results from ELGs and QSOs" (2026). In collaboration-wide review; expected to be submitted in early December 2026.

3. DESI Collaboration et al. (including **Q. Li**). "DESI DR2 Results III: The Mass of the Milky Way". arXiv:2611.05518 [astro-ph.GA] (2026).

4. DESI Collaboration, A. G. Adame, J. Aguilar, S. Ahlen, O. Alves, et al. (including **Q. Li**). "DESI DR2 Results IV: Alcock-Paczyński Measurements from the Lyman Alpha Forest and Cosmological Constraints". arXiv:2607.27410 [astro-ph.CO] (2026).

5. Herbold, M., A. Cuceu, P. Martini, H. K. Herrera-Alcantar, J. Guy, et al. (including **Q. Li**). "Validation of the DESI DR2 Lyα forest full-shape analysis". arXiv:2607.27411 [astro-ph.CO] (2026).

6. Ruiz-Herrera Bernal, M. F., S. Avila, A. Font-Ribera, H. K. Herrera-Alcantar, D. Alonso, et al. (including **Q. Li**). "CoLoRe-2LPT: Lyman-α mock catalogues for the validation of DESI cosmological analyses". arXiv:2607.27412 [astro-ph.CO] (2026).

7. Fernández-García, E., F. Prada, A. Smith, J. DeRose, A. J. Ross, et al. (including **Q. Li**). "DESI DR2 reference mocks: clustering results from UCHUU-BGS and LRG". *JCAP* 05, 002 (2026).

8. Liu, B., W. Luo, M. Kilbinger, S. Fu, I. Dell'Antonio, et al. (including **Q. Li**). "CFHT MegaCam Two Deep Fields Imaging Survey (2DFIS) I: Overview". arXiv:2602.08312 [astro-ph.GA] (2026).

9. Ratajczak, J., K. S. Dawson, N. Weaverdyck, J. Aguilar, S. Ahlen, et al. (including **Q. Li**). "The Compilation and Validation of the Spectroscopic Redshift Catalogs for the DESI-COSMOS and DESI-XMM-LSS Fields". *AJ* 171, 71 (2026).

10. Casas, L., H. K. Herrera-Alcantar, J. Chaves-Montero, A. Cuceu, A. Font-Ribera, et al. (including **Q. Li**). "Validation of the DESI DR2 Lyα BAO analysis using synthetic datasets". *Phys. Rev. D* 113, 023520 (2026).

11. Gwyn, S., A. W. McConnachie, J.-C. Cuillandre, K. C. Chambers, E. A. Magnier, et al. (including **Q. Li**). "UNIONS: The Ultraviolet Near-infrared Optical Northern Survey". *AJ* 170, 324 (2025).

12. Liu, H., Y. Lu, H. Hong, H. Wang, H. Mo, et al. (including **Q. Li**). "Radio AGN feedback sustains quiescence only in a minority of massive galaxies". arXiv:2511.06037 [astro-ph.GA] (2025).

13. Brodzeller, A., M. Wolfson, D. M. Santos, M. Ho, T. Tan, et al. (including **Q. Li**). "Construction of the damped Lyα absorber catalog for DESI DR2 Lyα BAO". *Phys. Rev. D* 112, 083510 (2025).

14. DESI Collaboration, M. Abdul Karim, J. Aguilar, S. Ahlen, C. Allende Prieto, et al. (including **Q. Li**). "DESI DR2 results. I. Baryon acoustic oscillations from the Lyman alpha forest". *Phys. Rev. D* 112, 083514 (2025).

15. Elbers, W., A. Aviles, H. E. Noriega, D. Chebat, A. Menegas, et al. (including **Q. Li**). "Constraints on neutrino physics from DESI DR2 BAO and DR1 full shape". *Phys. Rev. D* 112, 083513 (2025).

16. Garcia-Quintero, C., H. E. Noriega, A. de Mattia, A. Aviles, K. Lodha, et al. (including **Q. Li**). "Cosmological implications of DESI DR2 BAO measurements in light of the latest ACT DR6 CMB data". *Phys. Rev. D* 112, 083529 (2025).

17. Lodha, K., R. Calderon, W. L. Matthewson, A. Shafieloo, M. Ishak, et al. (including **Q. Li**). "Extended dark energy analysis using DESI DR2 BAO measurements". *Phys. Rev. D* 112, 083511 (2025).

18. Lin, S., W. Luo, Y. F. Cai, Q. Guo, L. Wei, et al. (including **Q. Li**). "The density profile of Milky Way dark matter halo constrained from the OGLE microlensing sky map". *A&A* 693, L8 (2025).

19. Mo, G., Q. Wang, X. Ren, W. Yan, **Q. Li**, et al. "Testing Cotton gravity as dark matter substitute with weak lensing". *Sci. China Phys. Mech. Astron.* 68, 240412 (2025).

20. Luo, W., J. D. Silverman, S. More, A. Goulding, H. Miyatake, et al. (including **Q. Li**). "Dark Matter Halos of Luminous Active Galactic Nuclei from Galaxy–Galaxy Lensing with the HSC Subaru Strategic Program". *ApJ* 977, 59 (2024).

21. Guo, Q., L. Wei, W. Luo, S. Lin, **Q. Li**, et al. "Static Microlensing: Concept, Method and Candidates". arXiv:2411.02161 [astro-ph.GA] (2024).

22. Zhang, Z., M. Kilbinger, F. H. Peters, **Q. Li**, W. Luo, et al. "Point spread function errors for weak lensing – density cross-correlations: Application to UNIONS". *A&A* 691, A75 (2024).

23. Zhang, Z., H. Wang, W. Luo, H. Mo, J. Zhang, et al. (including **Q. Li**). "Halo Mass-observable Proxy Scaling Relations and Their Dependencies on Galaxy and Group Properties". *ApJ* 960, 71 (2024).

Books (Translation)

1. Zee, A. *Chinese version of A. Zee's Quantum field theory in a nutshell*. Trans. by J. Cai, Y. He, J. Li, Z. Li, H. Liang, et al. (including **Q. Li**). Hefei, China: USTC Press, 2022. ISBN: 9787312051890.

Research Interests

Basically Observational Cosmology.

Dark Energy	Baryon Acoustic Oscillations, Lensing shear ratio, phenomenological dark energy models
Dark Matter	Gravitational Lensing, Modified Newtonian Dynamics, Dark Matter Substructures, Indirect Dark Matter Detection
Large-scale Structures	Galaxy Clustering, Galaxy groups and Clusters
Galaxy Evolution	Galaxy–Black Hole Co-evolution, Galaxy–Halo Connection

Research Mentoring

2026
- **Hanzhang Zhang, Logan McClure, and Chase Johansen**, undergraduate students at the University of Utah.
  Project: Selection function characterization for DESI Emission Line Galaxies.
- **Hanzhang Zhang**, undergraduate student at the University of Utah.
  Project: Color split tests for BAO measurements.

Talks & Presentations

Oct. 2026
- **Probing Evolution of Dark Energy through Galaxy Surveys**, *Waterloo Centre for Astrophysics Astroseminar, University of Waterloo*
- **Assessing the Uncertainty of DESI BAO Measurements**, *Cosmology Group Meeting, Perimeter Institute*

Jul. 2026
- **Plenary Talk: DR3 BAO Plans and Updates**, *DESI Collaboration Meeting at Durham*

Jun. 2026
- **Characterizing Cosmic Acceleration with Geometric Probes from Galaxy Surveys**, *Shanghai Jiao Tong University*
- **Characterizing Cosmic Acceleration with Geometric Probes from Galaxy Surveys**, *University of Science and Technology of China*
- **Characterizing Cosmic Acceleration with Geometric Probes from Galaxy Surveys**, *Sun Yat-sen University*

May 2026
- **Characterizing Cosmic Acceleration with Geometric Probes from Galaxy Surveys**, *Tsinghua University*

Apr. 2026
- **Kinematic Lensing Ratio: A New Geometric Probe of Dark Energy**, *Dark Energy from Home 2026*

Dec. 2025
- **DESI Spotlight Talk: BAO from DR2 and DR3**, *DESI Collaboration Meeting at NOIRLab*

Nov. 2025
- **Understanding the Uncertainty of Baryon Acoustic Oscillations**, *Graduate Research Seminar, University of Utah*

Jul. 2025
- **BAO Validation for Uchuu Reference Mock**, *DESI Collaboration Meeting at Berkeley*

Jun. 2025
- **DESI BAO and Evolving Dark Energy**, *Southern University of Science and Technology 3rd Student Physics & Mathematics Workshop, invited*

Mar. 2025
- **DESI DR2 BAO Results and Cosmological Implications**, *Graduate Student Journal Club, USTC, invited*

Dec. 2024
- **Introduction to Modern Cosmology**, *Southern University of Science and Technology 2nd Student Physics & Mathematics Workshop, invited*

Dec. 2023
- **Introduction to Modern Astrophysics**, *Southern University of Science and Technology 1st Student Physics & Mathematics Workshop, invited*

Jun. 2023
- **Stellar–Subhalo Mass Relation as a Probe to Galaxy–Subhalo Connection**, *China Gravitational Lensing Workshop XI, Qingdao*

Oct. 2022
- **Subhalo Properties in SDSS DR7 with Galaxy–Galaxy Lensing**, *Cluster Seminar at Department of Astronomy, University of Michigan*

Sep. 2022
- **Subhalo Properties from SDSS Weak Lensing**, *2022 PKU-DoA Undergraduate Astronomy Symposium at Kavli Institute for Astronomy and Astrophysics, Peking University*

Jun. 2022
- **Dark Matter Halo Mass Measurement with Weak Lensing**, *Undergraduate Academic Annual Conference, Department of Astronomy, USTC*

Science Collaborations

DESI — **Dark Energy Spectroscopic Instrument**, *Sponsored Member*
Co-lead of the BAO Topical Group (DR3 analysis). For the DR2 BAO analysis: consistency tests of the BAO measurements (key paper), galaxy-property split and post-unblinding tests (validation paper), and BAO validation of the Uchuu reference mocks (first-author paper).

UNIONS — **Ultraviolet Near-Infrared Optical Northern Survey**, *Member*
Led the weak lensing measurement of the black hole–halo mass relation (first-author paper), follow-up work in preparation.

XCS — **XMM Cluster Survey / X-ray Cluster Science**, *Member*
Led the weak-lensing mass calibration of X-ray cluster scaling relations

Honors & Awards

2024 Swigart First-Year Summer Graduate Research Fellowship, *University of Utah*

2023 Outstanding Graduate, *University of Science and Technology of China*

2022 Excellent Poster Prize, Undergraduate Research Symposium, *University of Science and Technology of China*

Linbridge Prize for Excellent Undergraduate Research Projects in Astronomy and Astrophysics, *Kavli Institute for Astronomy and Astrophysics, Peking University*

National Astronomical Observatory of China Fellowship

2021 Silver Prize, USTC Scholarship for Excellent Student, *University of Science and Technology of China*

2020 First Prize and the Best Record Winner, Astronomy Competition for College Students in South China, *Sun Yat-sen University*

2018 Silver Medal, China National Astronomy Olympiad, *Beijing Planetarium*

Teaching Experience

2024 SP (Discussion) Teaching Assistant — ASTR1060 *The Universe*

2023 FA Grader — ASTR3070 *Foundations of Astronomy*

2022 SP Teaching Assistant — *Observational Astronomy*

Non-Academic Service & Outreach

Jun. 2026– Graduate Research Seminar, Department of Physics and Astronomy, University of Utah, *Coordinator*

Oct. 2025– Graduate Student Association Committee, Department of Physics and Astronomy, University of Utah, *Officer*

Sep. 2021–Sep. 2022 'Questing For Truth' Natural Science Club, *Club President*

Dec. 2020–Sep. 2021 United Academic Forum of Basic Science for Undergraduates, *Council Member*

Sep. 2021 Popular Physics Lecture for Freshmen, *Organizer and Speaker*

Feb.–Mar. 2021 Online Seminar on General Relativity and Cosmology for Undergraduates, *Organizer and Speaker*

Jan. 2021 Lecture Series on Popular Astronomy at Huafu High School, *Lecturer*